

Johnson C. Smith University
College Algebra | MTH 131 | Sections B and I | Fall 2012

“Every act of conscious learning requires the willingness to suffer an injury to one’s self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all.”

–Thomas Szasz, author, professor of psychiatry (1920-)

Instructor Information

Name: Dr. James Ashe

Office location: 307 Davis Hall (back); office hours will also be held in the Math Lab, EDU 115

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Office phone: (704) 378-1037

Office hours: MWF 10–11am; MW 1–4pm; Tu 12pm–2pm; F 1pm–1:30pm, by appointment, or whenever I’m in my office.

Course Description

This is a 3 credit hour course designed to provide an investigative approach to college algebra using the computer and graphing calculator as tools. The topics covered include basic concepts of algebra, functions and graphs, polynomials and rational functions, exponential and logarithmic functions, systems of equations and solving systems of equations using matrices, sequences and series. At least one hour per week of computer assisted laboratory instruction is required. There are no prerequisites for this course.

Instructional methods will primarily consist of lectures and in-class discussions, however other methods may be adopted according to the particular needs of the class.

Course Materials

Textbook/MyMathLab: A textbook/Ebook and a MyMathLab access code are required. New textbooks and Ebooks usually come packaged with a MyMathLab code.

Textbook: *College Algebra*, 5th Edition, Robert Blitzer. ISBN: 0-1314-9264-0

Alternative options are available here:

<http://www.mypearsonstore.com/bookstore/product.asp?isbn=0321559835>

Calculator: TI-83, TI-83+, or TI-84+

Students should bring their calculator and textbook to class.

Grades

Grades will be taken from homework, quizzes, and Exams as follows:

Evaluation: Quizzes = 20%; Exams 1, 2, 3, 4, and the Midterm Exam = 12% each; Final Exam = 20%.

Grading scale: **A** $\geq$ 90 > **B** $\geq$ 80 > **C** $\geq$ 70 > **D** $\geq$ 60 > **F**

Quizzes: Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, MyMathLab assignments, take home assignments, and in-class projects. The lowest quiz grade will be dropped.

Final Exam and Midterm: Both the final and midterm exam will be common comprehensive exams given in-class; see the exam schedule jcsu.edu/happenings/exam-schedule. If beneficial, the final exam score will replace the lowest exam score.

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence.

Missed quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late take-home quizzes will have 25% deducted for each day it is late.

Missed exams: If a student misses an exam for a verifiable excuse, they should send me an e-mail *as soon as possible* so that a make-up can be arranged. Make-up exams may be significantly more difficult. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Course Content

Chapter 1: Equations and Inequalities

- 1.1 Graphs and Graphing Utilities
- 1.2 Linear Equations and Rational Equations
- 1.3 Models & Applications
- 1.4 Complex Numbers
- 1.5 Quadratic Functions
- 1.6 Other Types of Equations

Chapter 2: Functions and Graphs

- 2.1 Basics of Functions and Their Graphs
- 2.3 Linear Functions and Slope
- 2.4 More on Slope

Chapter 3: Polynomial and Rational Functions

- 3.1 Quadratic Functions
- 3.2 Polynomial Functions and Their Graphs

Chapter 4: Exponential and Logarithmic Functions

- 4.1 Exponential Functions
- 4.2 Logarithmic Functions
- 4.3 Properties of Logarithms

Important Dates

September 21 th	Last day to Add/Drop
October 5 th	Exam 1
October 26 th	Exam 2
November 2 nd	Midterm Exam
November 30 th	Exam 3
January 4 th	Exam 4
January 15 th –19 th	Common Final Exam

The above exam dates are subject to change. JCSU's Fall 2012 academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: jcsu.edu/happenings/exam-schedule